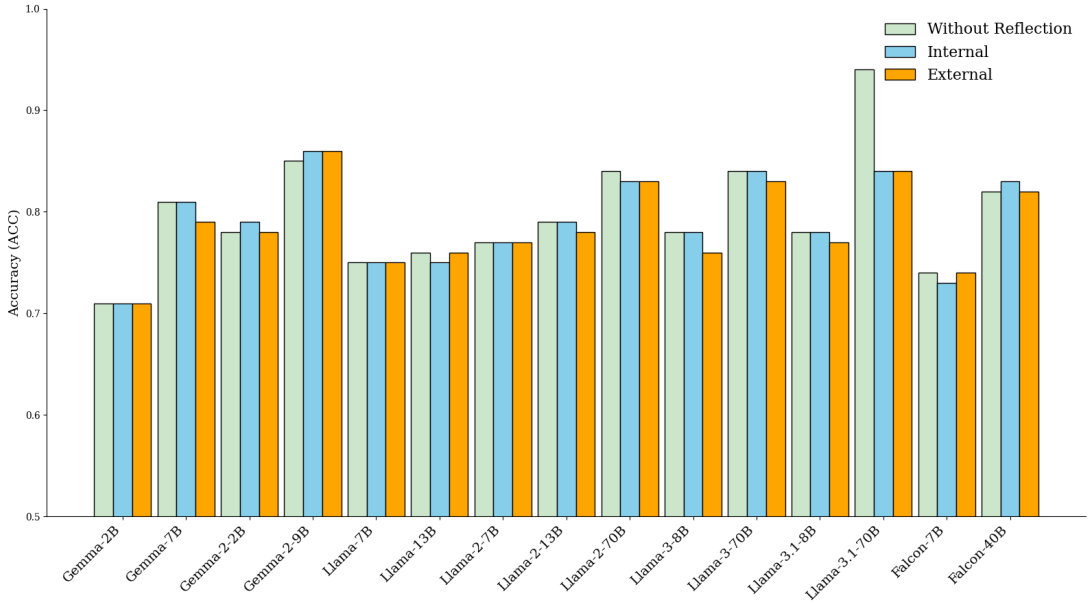

Figure 22: Accuracy (ACC) on ARC-E (general performance). LLMs fine-tuned with safety-reflection preserve general performance, comparable to standard fine-tuning.

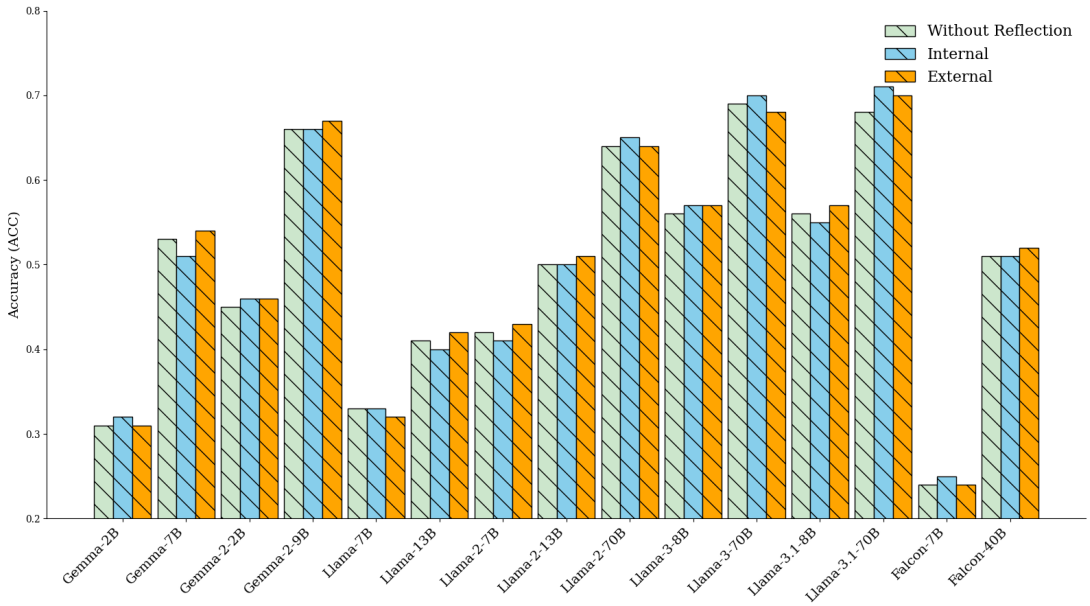

Figure 23: Accuracy (ACC) on MMLU (general performance). LLMs fine-tuned with safety-reflection preserve general performance, comparable to standard fine-tuning.

	Safety		Oversensitivity		General Performance		
	Xstest-H CR ↓	**Malicious** CR ↓	**Xstest-S** CR ↑	**OR-Bench** CR ↑	**MLLU** CR ↑	**GSM8K** CR ↑	**ARC-E** CR ↑
GEMMA1-2B							
Fine-Tuned w/o Rationale	0.10	0.07	0.74	0.68	0.31	0.12	0.71
Fine-Tuned w/ Internal Rationale	0.13	0.12	0.78	0.74	0.32	0.12	0.71
Fine-Tuned w/ External Rationale	0.12	0.07	**0.79**	**0.83**	0.31	0.12	0.71
GEMMA1-7B							
Fine-Tuned w/o Rationale	0.02	0.04	0.71	0.64	0.53	0.44	0.81
Fine-Tuned w/ Internal Rationale	0.05	0.01	0.82	0.71	0.51	0.42	0.81
Fine-Tuned w/ External Rationale	0.02	0.02	**0.90**	**0.75**	0.54	0.42	0.79
GEMMA-2-2B							
Fine-Tuned w/o Rationale	0.17	0.16	0.89	0.71	0.45	0.21	0.78
Fine-Tuned w/ Internal Rationale	0.20	0.20	0.90	0.80	0.46	0.21	0.79
Fine-Tuned w/ External Rationale	0.20	0.11	**0.91**	**0.73**	0.46	0.20	0.78
GEMMA-2-9B							
Fine-Tuned w/o Rationale	0.04	0.07	0.85	0.64	0.66	0.60	0.85
Fine-Tuned w/ Internal Rationale	0.07	0.10	0.88	0.68	0.66	0.61	0.86
Fine-Tuned w/ External Rationale	0.05	0.03	**0.90**	**0.72**	0.67	0.60	0.86
LLAMA1-7B							
Fine-Tuned w/o Rationale	0.01	0.01	0.69	0.48	0.33	0.09	0.75
Fine-Tuned w/ Internal Rationale	0.02	0.05	0.89	0.54	0.33	0.10	0.75
Fine-Tuned w/ External Rationale	0.02	0.01	**0.91**	**0.62**	0.32	0.10	0.75
LLAMA-2-7B							
Fine-Tuned w/o Rationale	0.02	0.03	0.74	0.74	0.41	0.13	0.76
Fine-Tuned w/ Internal Rationale	0.03	0.05	0.79	0.75	0.40	0.13	0.75
Fine-Tuned w/ External Rationale	0.03	0.04	**0.92**	**0.80**	0.42	0.13	0.76
LLAMA1-13B							
Fine-Tuned w/o Rationale	0.00	0.00	0.66	0.26	0.42	0.16	0.77
Fine-Tuned w/ Internal Rationale	0.02	0.03	0.80	0.37	0.41	0.16	0.77
Fine-Tuned w/ External Rationale	0.01	0.01	**0.89**	**0.41**	0.43	0.16	0.77
LLAMA-2-13B							
Fine-Tuned w/o Rationale	0.02	0.05	0.80	0.53	0.50	0.22	0.79
Fine-Tuned w/ Internal Rationale	0.04	0.07	0.92	0.58	0.50	0.21	0.79
Fine-Tuned w/ External Rationale	0.06	0.02	**0.97**	**0.63**	0.51	0.22	0.78
LLAMA-2-70B							
Fine-Tuned w/o Rationale	0.00	0.01	0.64	0.46	0.64	0.51	0.84
Fine-Tuned w/ Internal Rationale	0.00	0.03	0.92	0.64	0.65	0.51	0.83
Fine-Tuned w/ External Rationale	0.00	0.01	**0.96**	**0.68**	0.64	0.50	0.83
LLAMA-3-8B							
Fine-Tuned w/o Rationale	0.02	0.02	0.79	0.43	0.56	0.40	0.78
Fine-Tuned w/ Internal Rationale	0.02	0.05	0.88	0.53	0.57	0.37	0.78
Fine-Tuned w/ External Rationale	0.03	0.04	**0.92**	**0.63**	0.57	0.38	0.76
LLAMA-3-70B							
Fine-Tuned w/o Rationale	0.01	0.03	0.72	0.33	0.69	0.67	0.84
Fine-Tuned w/ Internal Rationale	0.02	0.04	0.84	0.58	0.70	0.70	0.84
Fine-Tuned w/ External Rationale	0.01	0.02	**0.88**	**0.50**	0.68	0.65	0.83
LLAMA-3.1-8B							
Fine-Tuned w/o Rationale	0.01	0.02	0.78	0.53	0.56	0.40	0.78
Fine-Tuned w/ Internal Rationale	0.04	0.02	0.85	0.58	0.55	0.38	0.78
Fine-Tuned w/ External Rationale	0.02	0.03	**0.92**	**0.68**	0.57	0.40	0.77
LLAMA-3.1-70B							
Fine-Tuned w/o Rationale	0.00	0.02	0.67	0.29	0.68	0.71	0.84
Fine-Tuned w/ Internal Rationale	0.01	0.04	0.75	0.43	0.71	0.72	0.84
Fine-Tuned w/ External Rationale	0.00	0.02	**0.86**	**0.46**	0.70	0.69	0.84
FALCON-7B							
Fine-Tuned w/o Rationale	0.05	0.03	0.69	0.61	0.24	0.04	0.74
Fine-Tuned w/ Internal Rationale	0.05	0.03	0.83	0.72	0.25	0.04	0.73
Fine-Tuned w/ External Rationale	0.06	0.04	**0.95**	**0.82**	0.24	0.05	0.74
FALCON-40B							
Fine-Tuned w/o Rationale	0.01	0.01	0.72	0.25	0.51	0.22	0.82
Fine-Tuned w/ Internal Rationale	0.00	0.04	0.86	0.42	0.51	0.22	0.83
Fine-Tuned w/ External Rationale	0.02	0.03	**0.93**	**0.51**	0.52	0.23	0.82

Table 3: Summary of Model Performance Across Three Evaluation Dimensions: False Refusal, Safety, and General Performance